

Asadullah Bhuiyan

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RESEARCH INTERESTS

Quantum dynamics: open and nonequilibrium quantum systems; dissipation and adaptive dynamics; topology, entanglement, purification, and boundary criticality.

Physics of learning: statistical physics of neural networks; prediction mechanisms and interpretability; effects of data structure, noise, and training on learning.

I combine analytical theory and numerical experiments to study how randomness, symmetry, and dynamical constraints shape information and collective behavior in quantum systems and neural networks.

EDUCATION

Cornell University

Aug 2021 - Jun 2027 (expected)

Ph.D. in Theoretical Physics. Advisor: Chao-Ming Jian.

Thesis in progress: *Random Processes in Topological Quantum Dynamics and Machine Learning*.

University of Alberta

2017 - 2021

B.Sc. in Engineering Physics, with distinction.

PUBLICATIONS AND PREPRINTS

Peer-reviewed journal articles

1. **A. Bhuiyan**, H. Pan, and C.-M. Jian. Free-Fermion Dynamics with Measurements: Topological Classification and Adaptive Preparation of Topological States. *Physical Review Research* **8**, 023147 (2026).
2. R. Boyack, **A. Bhuiyan**, A. Su, and F. Marsiglio. The bound-state solutions of the one-dimensional pseudoharmonic oscillator. *Journal of Mathematical Chemistry* **61**, 242-276 (2023).
3. **A. Bhuiyan** and F. Marsiglio. Edge Localized Schrödinger Cat States in Finite Lattices via Periodic Driving. *Physical Review B* **102**, 245302 (2020).
4. **A. Bhuiyan** and F. Marsiglio. Landau Levels, Edge States, and Gauge Choice in 2D Quantum Dots. *American Journal of Physics* **88**, 986-1005 (2020).
5. A. P. Kalra, S. D. Patel, **A. F. Bhuiyan**, J. Preto, K. G. Scheuer, U. Mohammed, J. D. Lewis, V. Rezania, K. Shankar, and J. A. Tuszynski. Investigation of the Electrical Properties of Microtubule Ensembles under Cell-Like Conditions. *Nanomaterials* **10**, 265 (2020).

Preprint

6. J. Tahmassebpur, **A. Bhuiyan**, H. Kim, and O. Lesser. Learning from almost nothing: How neural networks survive heavy input corruption. arXiv:2606.11319 (2026). Under review at *Physical Review Research*.

Manuscript in preparation

- A. Bhuiyan**, H. Pan, and C.-M. Jian. *Chiral critical state ensembles in topological adaptive quantum circuits*. In preparation.

Graduate Research Assistant, Cornell University

May 2022 - Present

Advisor: Chao-Ming Jian. Research in quantum dynamics and the physics of neural-network learning.

Neural-network learning under input corruption

- Initiated a study of how neural networks retain predictive information under heavy input corruption with intact training labels. Recruited two graduate students and a postdoc, organized research meetings, and contributed to the writing and structure of the paper.
- Developed and wrote a linear-response explanation within neural tangent kernel (NTK) theory: permutation symmetry near pure noise relates the leading class-dependent predictor to overlaps with centered class means.
- Guided a complementary high-noise expansion separating signal from fluctuations. Compared fitted response predictions with ensemble-mean outputs of wide networks trained with squared-error loss in Python/PyTorch, varying input noise and initialization independently.

Topological classification and adaptive quantum-state preparation

- Constructed proofs identifying which symmetry classes admit symmetry-preserving Gaussian measurements without post-selection in free-fermion dynamics, using group theory and spectral analysis. Performed the technical calculations and wrote the technical appendices.
- Developed an effective Lindbladian and closed covariance equations for measurement-and-feedback preparation of Chern-insulator states. The ideal construction with exponentially localized operations predicts an exponential relaxation timescale independent of system size, measured in protocol cycles.
- Carried out independent numerical checks and used covariance, transfer-matrix, and Choi representations to analyze Gaussian evolution and singular measurement limits. Contributed substantially to the main text and scientific figures.

Boundary criticality in adaptive quantum circuits | ongoing

- Investigating critical boundary-state ensembles in finite-range adaptive circuits, with the aim of characterizing chirality and preparing edge modes. Connecting topological-state preparation to continuum expectations for entanglement and charge fluctuations.
- Compared Rényi entropy, mutual information, and charge-fluctuation diagnostics across seven system sizes, using uncertainty estimates that account for correlations between measurements.

Monitored bosonic dynamics and stochastic field theory

- Proved an exponentially decaying upper bound on ensemble-averaged second Rényi entropy for all-to-all bosonic circuits with Haar-random passive Gaussian gates and single-mode heterodyne measurements, using covariance inequalities and Haar averaging.
- Derived entropy-replica Hamiltonians and spin-chain descriptions, including boundary states, and investigated continuum descriptions of monitored dynamics. Studied late-time covariance fluctuations through stochastic Riccati equations, perturbation theory, and Itô calculus.
- Identified Hilbert-space truncation artifacts in apparent critical scaling by varying initial states and simulation cutoffs in Python and Bosonic Qiskit.

Interacting edges in fractional topological insulators

- Calculated clean interacting-edge conductance using Kubo response in a fractional topological-insulator model. Used conformal perturbation theory to derive one-loop magnetic-impurity flow equations and temperature-dependent couplings for a diagonal velocity matrix.

Exploratory collaboration on neural quantum states

Initiated a collaboration exploring neural quantum states and learning methods for quantum systems.

EARLIER RESEARCH AND INDUSTRY EXPERIENCE

Undergraduate Research Assistant, University of Alberta

May 2018 - Aug 2021

- Conducted analytical and numerical research spanning driven quantum systems, quantum-dot physics, singular potentials, electron-phonon models, and microtubule biophysics.
- Studied edge-localized states in periodically driven finite lattices, explaining differences between continuum and tight-binding descriptions through boundary conditions and effective edge-site energies.
- Analyzed Landau levels and edge states in quantum dots, separating gauge-invariant densities and currents from numerical eigenvector ambiguity. Used a small gauge-invariant perturbation to resolve degeneracies.
- Coauthored analytical and numerical work on bound states of the one-dimensional pseudoharmonic oscillator, including singular-potential limits.
- Implemented an iterative MATLAB solver for Green's-function convolutions in the two-dimensional Hubbard-Holstein model to study competing superconducting and charge-density-wave instabilities.
- Contributed software, formal analysis, and validation to a study of the electrical properties of microtubule ensembles under cell-like conditions.

Product Engineer Intern, Microchip Technology

Sep - Dec 2019

Burnaby, British Columbia. Wrote Python scripts to automate microchip testing and generate spreadsheet reports of results.

TALKS

Dartmouth Quantum Nano Seminar

11 Nov 2025

Fermionic Monitored Dynamical Systems: Topological Classification & Adaptive Circuit Constructions. Dartmouth College. Invited seminar.

APS Global Physics Summit, Anaheim

19 Mar 2025

Topological Modes in Monitored Dynamics. Contributed talk. [Program](#).

APS March Meeting, Las Vegas

6 Mar 2023

Free Bosons Purify Faster than Free Fermions in 0D. [Program](#).

HONORS AND AWARDS

Cornell Graduate Fellowship

2025

NSERC Postgraduate Scholarship - Doctoral (PGS-D)

2022

ADVANCED TRAINING

Prospects in Theoretical Physics, Institute for Advanced Study

2023

Summer-school participant. Theme: *Understanding Confinement*.

Boulder School for Condensed Matter and Materials Physics

2022

Summer-school participant. Theme: *Hydrodynamics Across Scales*.

TEACHING EXPERIENCE

Seminar Instructor, Cornell University

Fall 2026

AEP 4230: *Statistical Physics*.

Graduate Teaching Assistant, Cornell University

Spring 2022

PHYS 2214: *Oscillations, Waves, and Quantum Physics*. Seminar and laboratory instruction.

Graduate Teaching Assistant, Cornell University

Fall 2021

PHYS 1201: *Why the Sky is Blue*. Seminar instruction.

MENTORING

Volunteer research mentorship | one year of weekly meetings

Mentored high-school student Samyak Jain through introductory quantum mechanics, analytical and computational single-particle problems, and qubits and quantum information, progressing to a Qiskit research project on competition between decoherence and entangling unitary dynamics. Provided weekly meetings and ongoing guidance between meetings.

His project, *Entanglement Sudden Death in Interacting Quantum Circuits*, received Third Place Awards at the Massachusetts State Science and Engineering Fair and Massachusetts Region IV Science and Engineering Fair in 2025.

Graduate-school application advising

Advised two senior undergraduates on program fit and selection of letter writers, and substantially edited their statements of purpose. One received offers from UC Berkeley and Princeton in experimental condensed-matter physics; the other received an offer from the University of Michigan in high-energy theory.

First-year cohort mentoring

Served as a mentor for first-year students.

OUTREACH

Moiré physics learning activity and teacher workshops

Co-developed with the research group an activity for middle-school students involving measurement of moiré periods and connections between the resulting patterns and quantum materials. Presented the activity at two Cornell-hosted workshops for teachers.

Physics demonstrations for K-12 students

Presented physics experiments directly to K-12 students as part of science outreach.

RESEARCH METHODS AND COMPUTING

Analytical methods: Gaussian quantum dynamics; covariance, transfer-matrix, and Choi formulations; open-system and Lindblad dynamics; stochastic processes and Itô calculus; perturbation theory and asymptotics; replica methods; bosonization, Kubo response, and renormalization-group calculations; neural tangent kernels and symmetry-based response theory.

Numerical methods: stochastic-trajectory ensembles, numerical linear algebra and integration, parameter sweeps, finite-size comparisons, entanglement and topology diagnostics, neural-network experiments, fitting, and uncertainty analysis.

Computing: Python (NumPy, SciPy, PyTorch, pandas, Matplotlib, Qiskit, and Bosonic Qiskit); MATLAB; Mathematica; LaTeX.

Research code: Noisy neural networks; Adaptive free-fermion circuits.